

Empirical Evaluation of the Resilience of Novel S-Box Implementations Against Power Side-Channel Attacks

Master's Degree in Computer Engineering

Candidate: Samuele Yves Cerini
Supervisor: Paolo Ernesto Prinetto



**Politecnico
di Torino**

Outline

1. **Side-Channel Power Analysis**
 - a. Definition and goals of the attack
2. **Countermeasures against SCA**
 - a. Rationale
 - b. Problems
 - c. Why S-Boxes?
3. **Thesis Contribution**
 - a. Motivations
 - b. Results
 - c. Conclusions and Future Work
4. **Q&A**

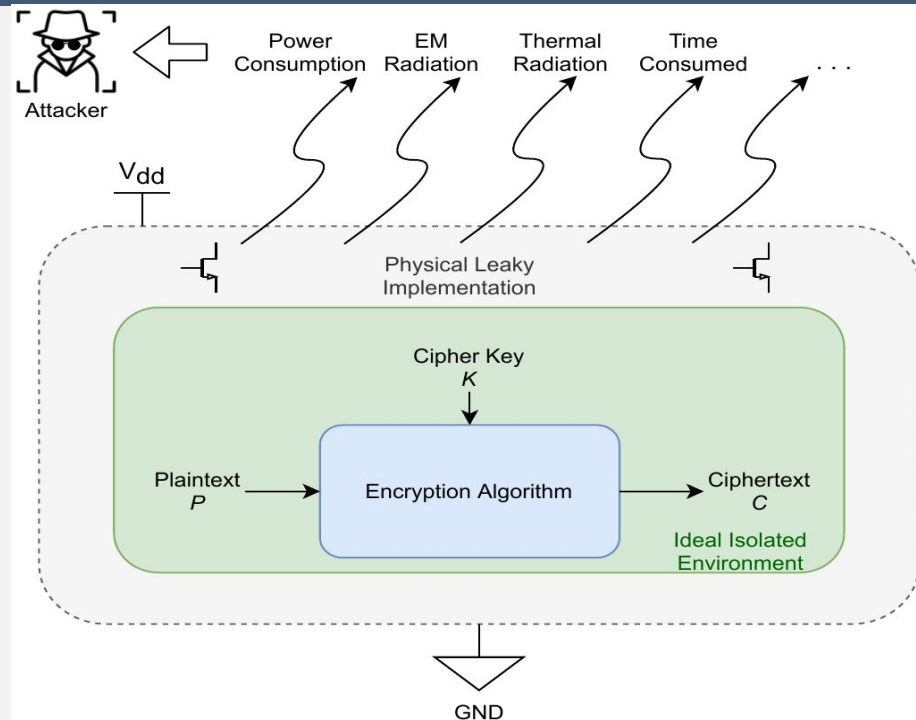
Side-Channel Power Analysis

Side-channel Analysis (SCA):

- Goal:
 - Retrieve sensitive data
 - cryptographic keys
 - Targets operating devices
 - IoT devices, embedded systems
- More efficient than classical cryptanalysis
 - More convenient for an attacker, less data needed

Power Analysis (PA):

- **Simple Power Analysis (SPA)**
 - Visual analysis
 - A single power trace may be sufficient
- **Differential/Correlation Power Analysis (DPA/CPA)**
 - Based on automated statistical computations
 - Multiple power traces are needed



Countermeasures against SCA - Why S-Boxes?

Physical emissions cannot be prevented, but...

1. Leakage can be reduced
 2. What cannot be prevented can be “scrambled”
- Problem:
 - **High impact on performance and code size**
 - Small *embedded devices* cannot afford such impact
 - Solution:
 - Move to the mathematical level, which is device-independent

Why target the S-Box structure?

- Crucial component for many block ciphers
- Look-Up-Table
 - Low memory impact (4x4 or 8x8)
- **Flexible**
- **Automated heuristic design is possible**

Thesis Contribution

Observation:

- **A plethora of novel S-Box have been proposed:**
 - designed with novel methodologies
 - pseudo-random generation
 - **chaos-based methods**
 - **heuristic methods**
 - many claim a theoretical resistance against SCA
 - most of these structures never left simulations
- **Theoretical metrics do exist:**
 - still in development
 - their effectiveness must be proven
 - Transparency Order
 - Improved Transparency Order
 - Confusion Coefficient

Motivations:

- **An empirical evaluation is needed:**
 - to benchmark the State of the Art of S-Box structures
 - to correlate the metrics with their claims

Analysis: S-Box structures

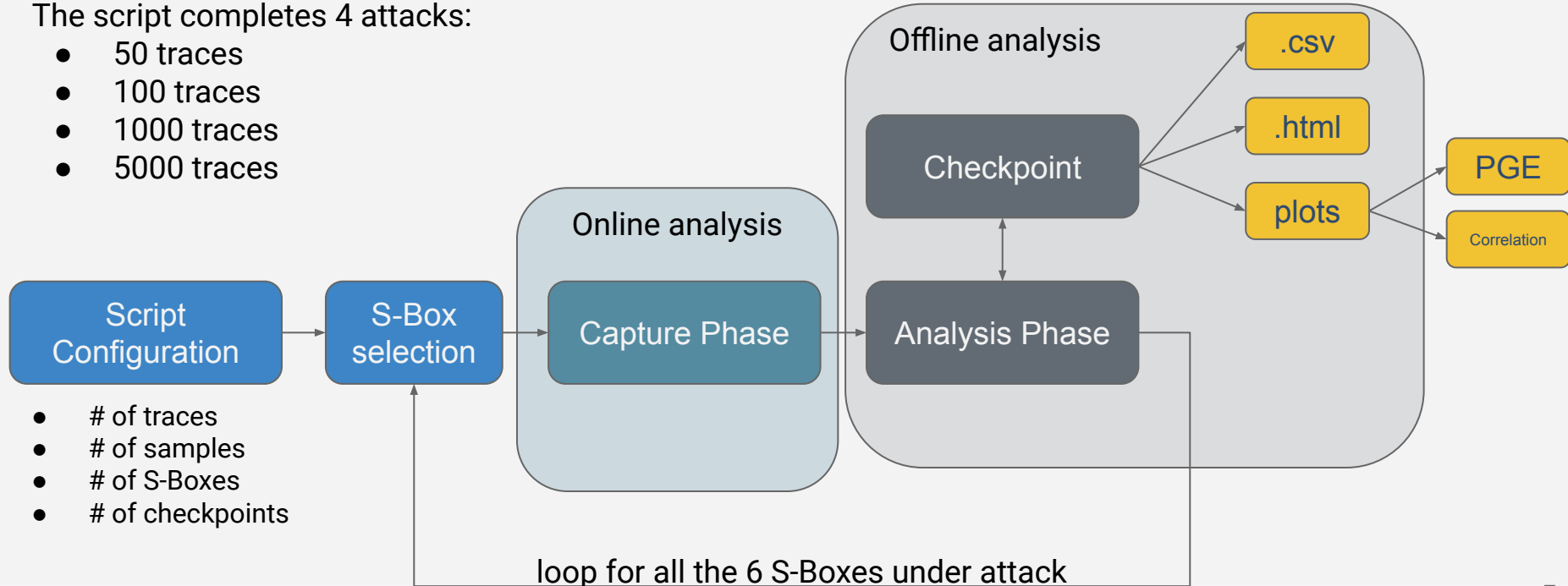
- Six different S-Box designs were tested:

S-Box	Nonlinearity (higher is better)	Differential Uniformity (lower is better)	Confusion Coeff. (CC) (higher is better)
AES (algebraic)	112	4/256	0.11
Freyre #1 (heuristic)	100	8/256	4.50
Freyre #2	100	8/256	4.49
Freyre #3	102	8/256	1.93
Hussain (chaos-based)	112	4/256	n.d.
Özkaynak	106.7	10/256	n.d.

Analysis: script overview

The script completes 4 attacks:

- 50 traces
- 100 traces
- 1000 traces
- 5000 traces

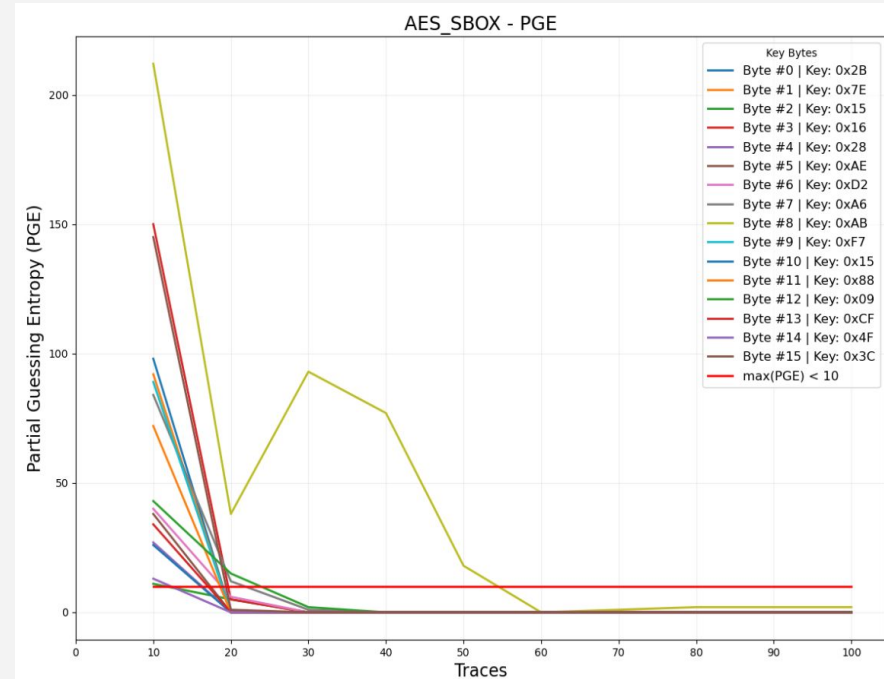
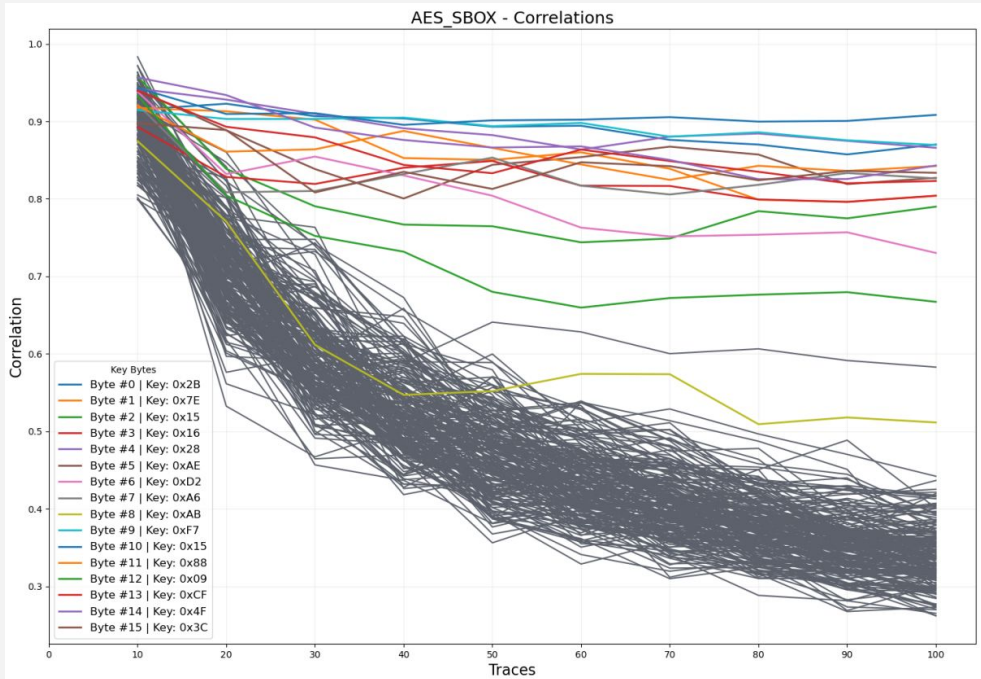


Results

Results can be grouped into 3 main sets:

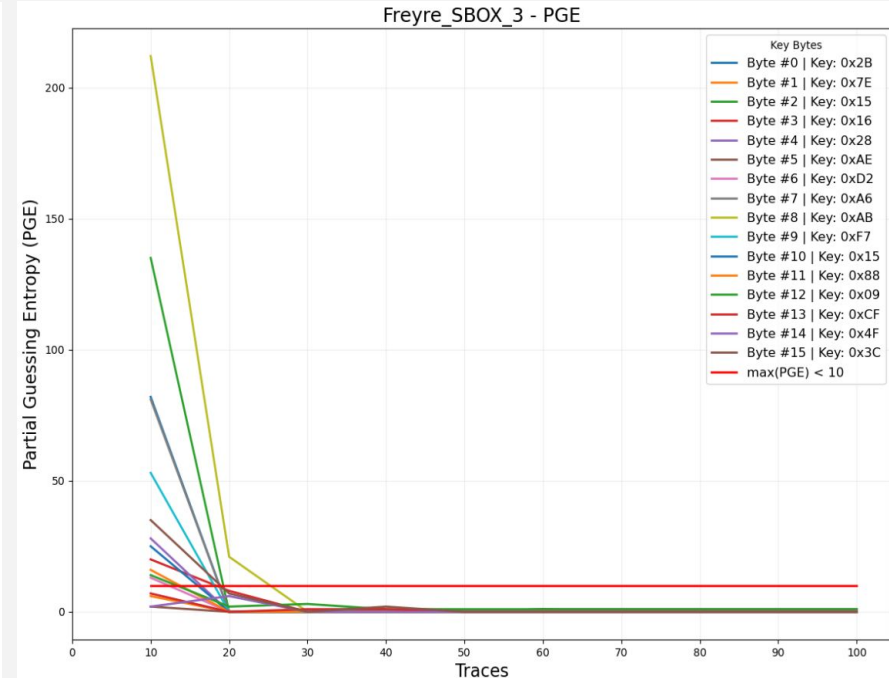
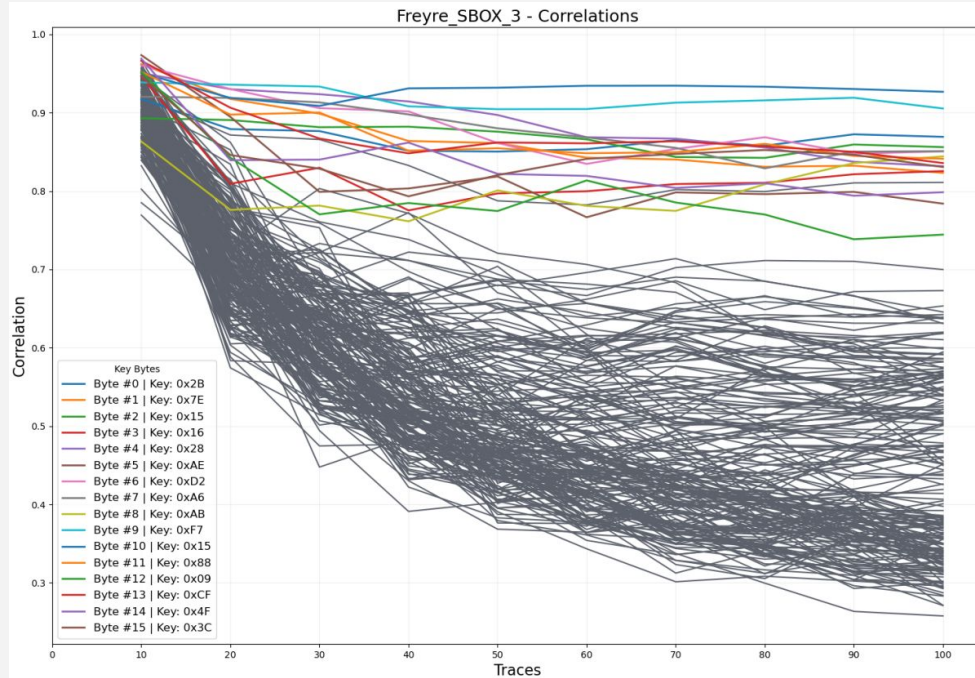
- **Set #1 (poor resistance against SCA)**
 - **AES S-Box**
 - Hussain S-Box
 - Özkaynak S-Box
- **Set #2 (moderate resistance against SCA)**
 - **Freyre S-Box #3**
- **Set #3 (good resistance against SCA)**
 - Freyre S-Box #1
 - **Freyre S-Box #2**

Results: 100 traces - AES S-Box



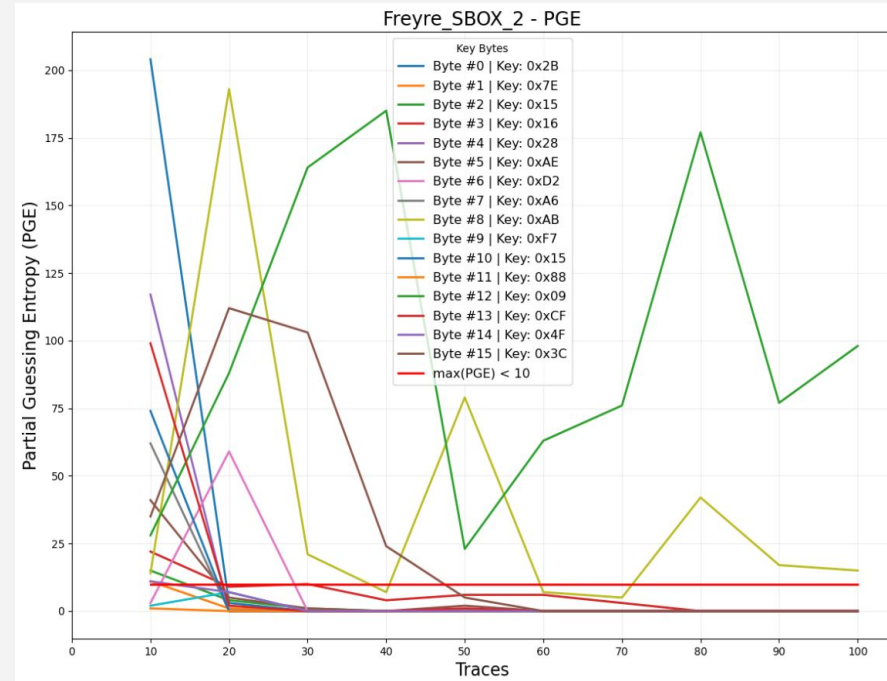
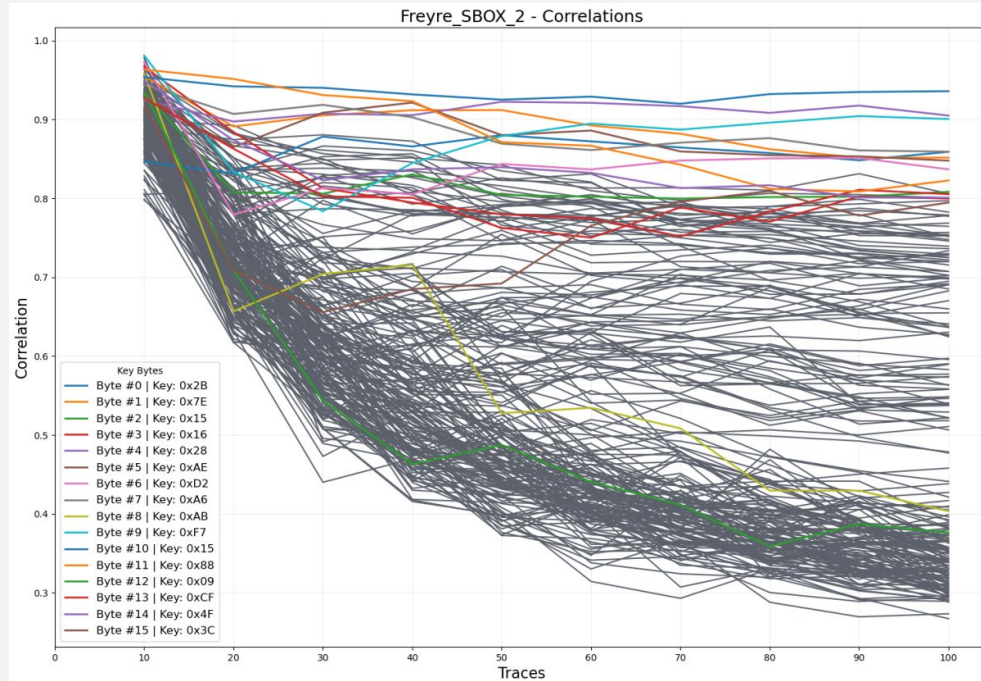
Confusion Coefficient: 0.11

Results: 100 traces - Freyre S-Box #3



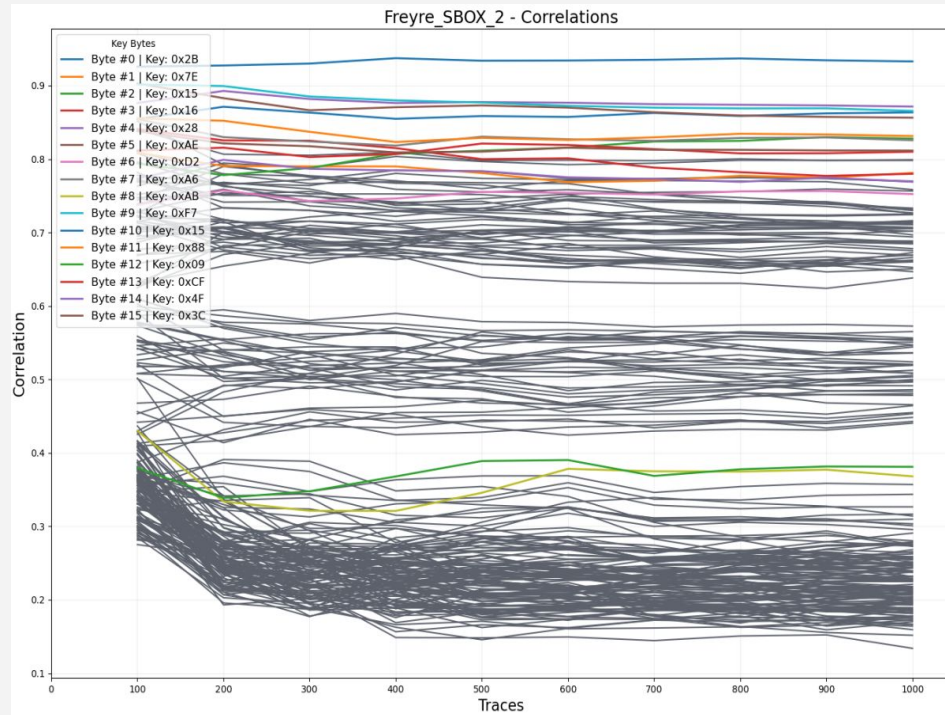
Confusion Coefficient: 1.93

Results: 100 traces - Freyre S-Box #2

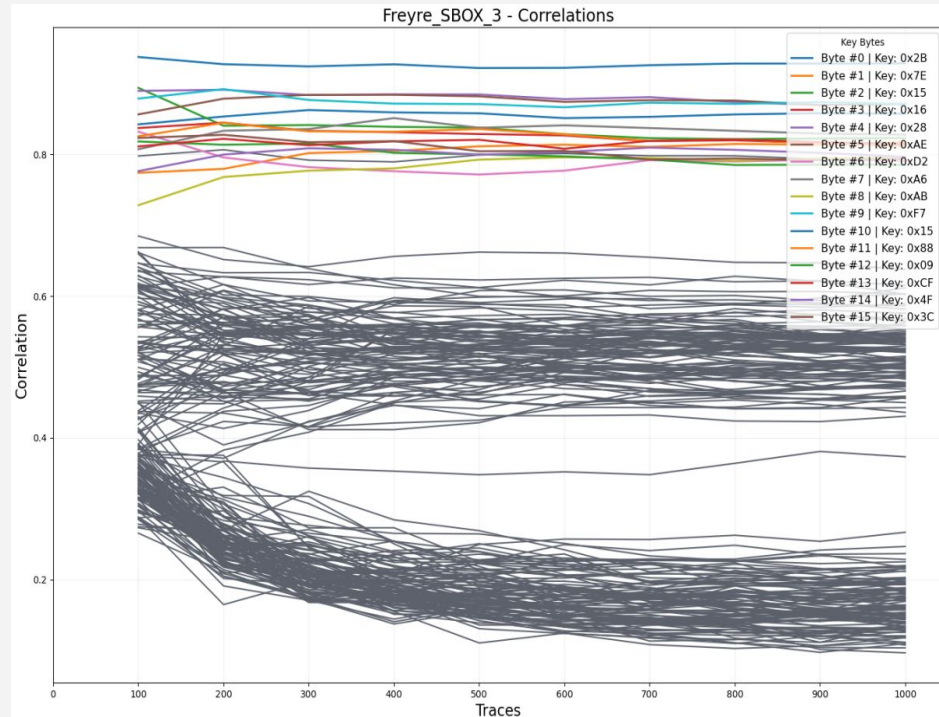


Confusion Coefficient: 4.49

Results: 1000 traces - Freyre S-Box #2 vs. #3



Confusion Coefficient: 4.49



Confusion Coefficient: 1.93

Conclusions, Future Work

Resistance to Side-Channel Analysis (SCA):

- **S-Boxes w/ CC = 5:**
 - **impressive results**
 - **# of Traces: increased by a factor 4 ÷ 10**
- **S-Boxes w/ CC = 2:**
 - results are not satisfactory enough
 - **# of Traces: no real improvements w.r.t. AES S-Box**
- **S-Boxes w/ no design criteria against SCA:**
 - poor results, as predicted
 - **# of Traces: 30-40 synchronously captured**

Future Work:

- Define empirical metrics to benchmark S-Boxes
- Relate other theoretical metrics to empirical results
- Repeat the analysis with a different device

Thank You!



**Politecnico
di Torino**

Q&A

**Empirical Evaluation of the Resilience of Novel S-Box
Implementations Against Power Side-Channel Attacks**

Candidate: Samuele Yves Cerini
Supervisor: Paolo Ernesto Prinetto